

Question ID: 441f0505

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	Medium

Question

While researching a topic, a student has taken the following notes:

- A lever is a simple machine consisting of a rigid beam and a fulcrum.
- The fulcrum is the point about which the beam pivots.
- The input force (effort) is the force applied to the lever.
- The output force (load) is the force that the lever exerts on another object.
- In first-class levers, the fulcrum is located between the effort and the load.
- In second-class levers, the load is located between the effort and the fulcrum.

The student wants to contrast first-class levers and second-class levers. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. In levers, the effort is the force applied to the lever; the load, in contrast, is the force that the lever exerts on another object.
- B. In first-class and second-class levers, the fulcrum and the load are in different locations.
- C. First-class levers are simple machines consisting of a rigid beam and a fulcrum, but then again, the same is true of second-class levers.
- D. In first-class levers, the fulcrum is located between the effort and the load, but in second-class levers, the load is located between the effort and the fulcrum.

Correct Answer: D

Rationale

Choice D is the best answer. The sentence contrasts first-class levers and second-class levers, explaining that the fulcrum in a first-class lever is between the effort and the load, whereas in a second-class lever the load is between the effort and the fulcrum.

Choice A is incorrect. The sentence defines two terms associated with levers; it doesn't contrast first-class levers and second-class levers. Choice B is incorrect. While the sentence seems to acknowledge a general difference in fulcrum and load locations between first-class and second-class levers, it does not specify what this difference is. Moreover, the sentence could be read as emphasizing a similarity—that in both types of levers, the fulcrum and load are in different locations. The sentence thus fails to effectively contrast the two types of levers. Choice C is incorrect. The sentence describes a similarity between first-class and second-class levers; it doesn't contrast them.